

Week-2 Day-1 : Contact Manager Application using TypeScript

Program:

```
// 1. Define Contact Interface
interface Contact {
  id: number;
  name: string;
  email: string;
  phone: string;
}

// 2. ContactManager Class
class ContactManager {
  private contacts: Contact[] = [];

  // Add a new contact
  addContact(contact: Contact): void {
    this.contacts.push(contact);
    console.log("Contact added successfully");
  }

  // View all contacts
  viewContacts(): Contact[] {
    return this.contacts;
  }

  // Modify an existing contact
  modifyContact(id: number, updatedContact: Partial<Contact>): void {
    let found = false;

    for (let i = 0; i < this.contacts.length; i++) {
      if (this.contacts[i].id === id) {
        if (updatedContact.name !== undefined) {
          this.contacts[i].name = updatedContact.name;
        }
        if (updatedContact.email !== undefined) {
          this.contacts[i].email = updatedContact.email;
        }
        if (updatedContact.phone !== undefined) {
          this.contacts[i].phone = updatedContact.phone;
        }

        found = true;
        console.log(" Contact modified successfully");
        break;
      }
    }

    if (!found) {
      console.error(" Error: Contact does not exist");
    }
  }
}
```

```

    }
}

// Delete a contact
deleteContact(id: number): void {
    let index = -1;

    for (let i = 0; i < this.contacts.length; i++) {
        if (this.contacts[i].id === id) {
            index = i;
            break;
        }
    }

    if (index === -1) {
        console.error(" Error: Contact does not exist");
        return;
    }

    this.contacts.splice(index, 1);
    console.log(" Contact deleted successfully");
}
}

// 3. Testing Script
const manager = new ContactManager();

// Add contacts
manager.addContact({
    id: 1,
    name: "Vijay",
    email: "vijay@gmail.com",
    phone: "9876543210"
});

manager.addContact({
    id: 2,
    name: "Arjun",
    email: "arjun@gmail.com",
    phone: "9123456789"
});

// View contacts
console.log("📁 All Contacts:");
console.log(manager.viewContacts());

// Modify contact
manager.modifyContact(1, {
    phone: "9999999999"
});

```

```

// Delete contact
manager.deleteContact(2);

// Try modifying non-existing contact
manager.modifyContact(3, {
  name: "Unknown"
});

// Final contact list
console.log("Final Contacts:");
console.log(manager.viewContacts());

```

Terminal Output:

```

TS contactManager.ts > ...
1 // Week-2 Day-1 : Contact Manager Application using TypeScript
2
3 // 1. Define Contact Interface
4 interface Contact {
5   id: number;
6   name: string;
7 }
8
9 // 2. Define Contact Manager Class
10 class ContactManager {
11   private contacts: Contact[] = [];
12
13   // Add a new contact
14   addContact(name: string, email: string, phone: string): void {
15     const contact: Contact = {
16       id: this.contacts.length + 1,
17       name,
18       email,
19       phone
20     };
21     this.contacts.push(contact);
22   }
23
24   // Modify an existing contact
25   modifyContact(id: number, name: string, email: string, phone: string): void {
26     const contact = this.contacts.find(c => c.id === id);
27     if (contact) {
28       contact.name = name;
29       contact.email = email;
30       contact.phone = phone;
31     } else {
32       console.error('Contact does not exist');
33     }
34   }
35
36   // Delete a contact
37   deleteContact(id: number): void {
38     const index = this.contacts.findIndex(c => c.id === id);
39     if (index !== -1) {
40       this.contacts.splice(index, 1);
41     } else {
42       console.error('Contact does not exist');
43     }
44   }
45
46   // View all contacts
47   viewContacts(): Contact[] {
48     return this.contacts;
49   }
50 }
51
52 // 3. Create an instance of ContactManager
53 const manager = new ContactManager();
54
55 // 4. Add some contacts
56 manager.addContact('Vijay', 'vijay@gmail.com', '9876543210');
57 manager.addContact('Arjun', 'arjun@gmail.com', '9123456789');
58
59 // 5. Modify a contact
60 manager.modifyContact(1, 'Vijay', 'vijay@gmail.com', '9999999999');
61
62 // 6. Delete a contact
63 manager.deleteContact(2);
64
65 // 7. View all contacts
66 console.log(manager.viewContacts());

```

```

web development/Wipro/TypecriptAssignment master x
• tsc contactManager.ts

web development/Wipro/TypecriptAssignment master x
• node contactManager.js
Contact added successfully
Contact added successfully
■ All Contacts:
[
  {
    id: 1,
    name: 'Vijay',
    email: 'vijay@gmail.com',
    phone: '9876543210'
  },
  {
    id: 2,
    name: 'Arjun',
    email: 'arjun@gmail.com',
    phone: '9123456789'
  }
]
Contact modified successfully
Contact deleted successfully
Error: Contact does not exist
Final Contacts:
[
  {
    id: 1,
    name: 'Vijay',
    email: 'vijay@gmail.com',
    phone: '9999999999'
  }
]

```